

1. Who is a Python Developer?

A Python Developer builds backend services, APIs, automation tools, and software systems using Python.

2. Responsibilities

Design applications, build APIs, integrate databases, write tests, automate workflows, deploy services, and maintain production systems.

3. Required Skills

Mandatory

Python • SQL • Git • FastAPI (or Django/Flask) • REST APIs • Debugging • Basic Testing

Preferred

Docker • Redis • Cloud (AWS/Azure/GCP) • CI/CD • System Design • Message Queues • Monitoring

4. Learning Roadmap

Phase 1

Goal: Master Python fundamentals

Estimated Time: ~1 month

Outcome: Build CLI applications and understand core syntax.

Phase 2

Goal: Learn OOP, Git, SQL, and data structures

Estimated Time: ~1–2 months

Outcome: Build database-backed applications.

Phase 3

Goal: Backend development with FastAPI/Django

Estimated Time: ~2 months

Outcome: Develop REST APIs with authentication.

Phase 4

Goal: Deployment and DevOps

Estimated Time: ~1–2 months

Outcome: Containerize, deploy, and monitor applications.

5. Projects

Beginner: Calculator, File Organizer, Weather App.

Skills Demonstrated: Python, file handling, APIs.

Intermediate: Blog API, Authentication Service, URL Shortener.

Skills Demonstrated: FastAPI, PostgreSQL, authentication, testing.

Advanced: Production REST API, Microservices Platform, Task Queue.

Skills Demonstrated: Docker, Redis, CI/CD, cloud deployment, scalability.

6. Resume Optimization Tips

Highlight measurable achievements, deployed projects, GitHub links, categorized skills, and quantified business impact.

7. Common ATS Mistakes

Missing keywords, poor formatting, generic summaries, no measurable achievements, missing portfolio links.

8. Interview Topics

Junior

Python syntax, OOP, functions, exceptions, SQL basics, Git.

Intermediate

Decorators, generators, REST APIs, authentication, Docker, testing.

Senior

System design, scalability, caching, microservices, cloud architecture, performance optimization.

9. Common Industry Tech Stack

Python, FastAPI, PostgreSQL, Redis, Docker, GitHub Actions, Nginx, AWS, pytest.

10. Career Progression

Junior Python Developer

Expected Skills: Python, Git, SQL, debugging.

Python Developer

Expected Skills: FastAPI/Django, REST APIs, databases, testing.

Senior Python Developer

Expected Skills: Docker, microservices, system design, cloud, mentoring, performance tuning.

Tech Lead / Architect

Expected Skills: Architecture, leadership, distributed systems, security, strategic technical decisions.